

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.7: Volume and Capacity — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.7: Volume and Capacity
Driving Question	DRIVING QUESTION: Why do the tea farmers' containers — a cone, a cylinder and a bucket — look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Predict and Anchor the Phenomenon: The Three Kericho Containers	We observed three containers from Mama Chebet's farm in Kericho: a cylindrical drum (A), a conical sisal cover (B) and a metal frustum bucket (C), all about 60 cm tall with a 42 cm opening (21 cm radius), though their shapes are different. We recorded our predictions of which holds most.	We learned that volume is the space inside a solid (cm^3 , m^3) and capacity is the amount a container can hold (mL, L), with $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$ and $1 \text{ m}^3 = 1000 \text{ L}$. We learned that equal height and opening do not tell us equal capacity.	We cannot yet explain why the containers hold different amounts, but we predict that shape matters. We have posted our questions on the Driving Question Board and will use mathematics to test our predictions in the next lessons.
Lesson 2 Cone versus Cylinder: Why Does the Cover Hold Only One Third of the Drum?	We filled a model cone to the brim and poured it into a cylinder of the same base radius and height. The cylinder was full after exactly 3 cone-fuls, in every trial and every group. For the Kericho containers, drum A holds 83.16 L and cover B holds 27.72 L, a ratio of exactly 3.	The volume of a cone is one third of the volume of a cylinder with the same base radius and height: $V = \frac{1}{3} \pi r^2 h$. With $\pi = \frac{22}{7}$, $r = 21 \text{ cm}$, $h = 60 \text{ cm}$, $V = 27,720 \text{ cm}^3$, and $1000 \text{ cm}^3 = 1 \text{ L}$ gives a capacity of 27.72 L.	The conical cover (B) and cylindrical drum (A) look alike because both are 60 cm tall with a 42 cm opening, but the cone narrows to a point, so it holds exactly one third of the drum: 27.72 L against 83.16 L. We still need to find out why the bucket (C) holds an amount in between.
Lesson 3 Volume of Pyramids: Does the One-Third Pattern Hold Beyond Cones?	We observed that a hollow square-based pyramid (base 30 cm x 30 cm, height 40 cm) filled a cuboid of the same base and height exactly 3 times, so the cuboid held $36,000 \text{ cm}^3$ and the pyramid $12,000 \text{ cm}^3$.	We learned that the volume of a pyramid is $V = \frac{1}{3} \times \text{base area} \times \text{perpendicular height}$, and that $12,000 \text{ cm}^3$ is 12 L. The same one-third relationship we found for the cone and cylinder holds for pyramids and cuboids.	We can explain that a pyramid holds one third of the cuboid with the same base and height, just as Mama Chebet's cover B holds one third of drum A. This moves us toward explaining the bucket (a frustum), which we will investigate next.
Lesson 4 Frustum of a Cone:	We measured bucket C: top radius $R = 21 \text{ cm}$, bottom radius $r = 14 \text{ cm}$ and	The volume of a frustum of a cone is $V(\text{large cone}) - V(\text{small cone}) = \frac{1}{3} \pi h (R^2$	The cover B (27.72 L), bucket C (58.52 L) and drum A (83.16 L) are all about 60 cm

Cutting the Bucket from a Cone	perpendicular height $h = 60$ cm. By extending the slanting sides we found a small cone ($h = 120$ cm) on top of the bucket and a large cone ($h = 180$ cm), using the ratio $r : R = 2 : 3$. Large cone = $83,160 \text{ cm}^3$, small cone = $24,640 \text{ cm}^3$.	$+ Rr + r^2$). Similar triangles give the missing height of the small cone. Using $\pi = 22/7$, bucket C holds $58,520 \text{ cm}^3 = 58.52 \text{ L}$.	tall with a 42 cm opening, but they hold different amounts because of shape. The cover tapers to a point, the bucket tapers only to a 28 cm base, and the drum does not taper. So the bucket holds more than the cover and less than the drum (about 70% of the drum). The same large-minus-small reasoning can be used to design a frustum container of any required capacity.
Lesson 5 Volume of a Frustum of a Pyramid: Designing the Maize Bin	We observed that a square-based maize bin model with top side 60 cm, bottom side 30 cm and height 40 cm gives $84,000 \text{ cm}^3$ both by subtracting the small pyramid ($12,000 \text{ cm}^3$) from the large pyramid ($96,000 \text{ cm}^3$) and by the formula with $A_1 = 3,600 \text{ cm}^2$, $A_2 = 900 \text{ cm}^2$ and $\sqrt{A_1 A_2} = 1,800 \text{ cm}^2$.	We learned that the volume of a frustum of a pyramid is $V = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 A_2})$, that $84,000 \text{ cm}^3 = 84 \text{ L}$, and that maize has a bulk density of $0.75 \text{ kg per litre}$, so the bin holds $84 \times 0.75 = 63 \text{ kg}$ of maize.	We can explain that, like Mama Chebet's bucket (C) as a large cone minus a small cone, a pyramid frustum is a large pyramid minus a small pyramid, which is why its volume lies between those of the small and large pyramids; we can use this to design a frustum container of the capacity a farmer needs.
Lesson 6 Problem Solving with Real Containers: Turkana Tank, Githeri Pot and Frustum Cups	We observed that three new Kenyan problems (a conical Turkana tank, a frustum githeri pot and a frustum cup) can each be solved by identifying the solid and applying the matching formula: the Turkana tank has $V = 6.16 \text{ m}^3 = 6,160 \text{ L}$, the githeri pot has $V = 80,520 \text{ cm}^3 = 80.52 \text{ L}$ (enough for 100 people at 0.5 L each, which needs 50 L), and each cup has $V = 462 \text{ cm}^3$.	We learned to choose the correct formula from the shape of the solid, to convert units (cm^3 to L , m^3 to L) before answering, to check answers for sense, and to divide one capacity by another to find a count: drum A holds $83,160 / 462 = 180$ cups.	We can now explain that the same mathematics that makes drum A (83.16 L) hold three times cover B (27.72 L) and bucket C hold 58.52 L also lets us size tanks, pots and cups, and design a frustum container of any capacity we need, because each container's capacity depends on its shape, its dimensions and the formula that matches it.
Lesson 7 Project: Model Making and Measurement – Building and Testing Our Own Kericho-Style Containers	What we observed: our cone, frustum of a cone and frustum of a pyramid models held measured amounts of water or sand that were close to, but usually a little different from, the volumes we calculated (record your own numbers: calculated cm^3 , measured mL , error $\text{ } \%$).	What we learned: the formulas $V = \frac{1}{3} \pi r^2 h$, $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$ and $V = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 A_2})$ predict the real capacity of containers within about $5\text{-}10\%$, and percentage error helps us judge how good the model is and why it differs.	What we can now explain: the Kericho containers differ in capacity because their shapes differ, and mathematics lets us calculate and design a container of the capacity we need. Differences between calculated and measured values come from things like card thickness, imperfect seams and shapes, and spilled water or sand.
Lesson 8 Culmination and Assessment: Explaining the Kericho Containers	We observed that the Kericho drum (A), conical cover (B) and metal bucket (C) all have a height of about 60 cm and an opening radius of 21 cm, yet hold 83.16 L , 27.72 L and 58.52 L respectively. We also observed the project models, whose measured capacities came within about $5\text{-}10\%$ of the calculated ones.	We learned that shape decides volume: the cone is exactly one third of the cylinder, $V = \frac{1}{3} \pi r^2 h$, and the pyramid is one third of its matching cuboid. A frustum is a large cone or pyramid with a smaller one removed, which gives $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$ and $V = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 A_2})$. We learned how to convert cm^3 to litres and how to calculate percentage error.	We can explain that the three containers look alike because they share the same height and opening, but the drum holds exactly 3 times the cover (83.16 L against 27.72 L). The bucket holds 58.52 L , between them, because it is the large cone ($83,160 \text{ cm}^3$) minus the small cone ($24,640 \text{ cm}^3$). Using the same reasoning for pyramids and frustums, we can design a frustum container that holds exactly the

			capacity needed, which is why volume and capacity knowledge is useful to Kericho tea farmers and many others.
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ATTRIBUTION AND LICENSING

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